

HANDS-ON LAB GUIDE · DAY 2

Reporting, Dashboards & Administration

Splunk Fundamentals · Day 2 — Hands-On Lab Guide

- Lab 8 — Statistical searches with stats and timechart
- Lab 9 — Application performance and error trends
- Lab 10 — Enrich data using CSV lookups
- Lab 11 — Create and schedule a report
- Lab 12 — Interactive operational dashboard
- Lab 13 — Threshold-based alerts
- Lab 14 — Troubleshoot a data-ingestion issue
- Lab 15 — Administration essentials

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Running scenario: **NovaMart** (fictional e-commerce retailer)
Applies to Splunk Enterprise 9.x / 10.x & Splunk Cloud Platform
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Contents

Before You Begin	3
Build Statistical Searches Using stats and timechart	5
Task 8.1 — Several functions in one stats	5
Task 8.2 — count, count(field) and dc	5
Task 8.3 — Group by region	5
Task 8.4 — chart: two dimensions	6
Task 8.5 — timechart	6
Task 8.6 — Time-based analysis with strftime and bin	7
Task 8.7 — top and rare	7
Task 8.8 — where	7
Task 8.9 — eval with stats: error rate per host	8
Analyse Application Performance and Error Trends	9
Task 9.1 — Checkout response time by payment method	9
Task 9.2 — Gateway latency: the week and the incident	9
Task 9.3 — Error and warning trend by day	10
Task 9.4 — Web error trend	10
Task 9.5 — Checkout success rate per day	10
Task 9.6 — Write up your findings	11
Enrich Data Using CSV Lookups	12
Task 10.1 — Upload the lookup files	12
Task 10.2 — Create lookup definitions	12
Task 10.3 — Revenue against target	13
Task 10.4 — Network zones for SSH logins (CIDR)	13
Task 10.5 — Latency against SLA	13
Task 10.6 — Filter out bots with inputlookup (subsearch)	13
Task 10.7 — Automatic lookup	14
Task 10.8 — Build a table with outputlookup, then use it	14
Create and Schedule a Report	16
Task 11.1 — Report 1: daily revenue by category (chart)	16
Task 11.2 — Report 2: payment gateway performance (formatted table)	16
Task 11.3 — Schedule Report 1 with a verifiable action	17
Task 11.4 — Export and share	17
Task 11.5 — Add a report to a dashboard	17
Build an Interactive Operational Dashboard	19
Part A — Start live traffic	19
Part B — Create the dashboard and inputs	19
Part C — KPI single values (top row)	20
Part D — Charts	20
Part E — Drilldown: click a status to see its events	20
Part F — A live panel	21
Part G — Arrange, save and share	21
Configure a Threshold-Based Alert	22
Part A — Start the webhook receiver (Terminal 2)	22
Part B — Backtest the alert logic on last week's data	22

Part C — Build the live alert search	22
Part D — Save it as a throttled alert with actions	23
Part E — Trigger it (Terminal 3)	23
Part F — A per-result brute-force alert	23
Part G — Troubleshoot and manage	24
Troubleshoot a Basic Data-Ingestion Issue	25
Part A — Deploy the vendor's configuration as delivered	25
Part B — Symptom: "there's no data"	25
Part C — Check the timestamps before you index anything	26
Part D — Fix all three faults in configuration files	26
Part E — Re-read the file: reset the fishbucket	27
Part F — Validate	27
Administration Essentials	29
Part A — Explore \$SPLUNK_HOME	29
Part B — Retention for novamart_inv	29
Part C — Create a restricted role and a user	29
Part D — Prove the restrictions	30
Part E — Monitoring Console tour	30
Part F — Admin searches in _internal and _audit	31
Wrap-up — stop the lab processes	31
Challenge Exercises	32
Appendix A: Answers and Solutions	33
Lab 8	33
Lab 9: model SRE summary	33
Lab 10	33
Lab 11	33
Lab 14: root-cause summary	33
Lab 15	34
Challenge solutions	34
Appendix B: Lab Troubleshooting	36
Appendix C: NovaMart Operations Dashboard (Classic Simple XML)	37
Appendix D: Expected Results Card (14–20 Sep 2026)	40

Before You Begin

These are the hands-on exercises for **Day 2 — Reporting, Dashboards and Basic Administration**. They continue from Day 1 (Labs 0–7) and use the same NovaMart data in `index=novamart`. Each lab maps to a module in the Day 2 *Participant Study Guide*. Every task shows its expected result.

Lab map

Lab	Title	Module	Time	You will...
8	Build statistical searches using stats and timechart	6	35 min	stats, chart, timechart, top, rare, where, eval with stats
9	Analyse application performance and error trends	6	30 min	Latency, response time, error trends, success rates, findings
10	Enrich data using CSV lookups	7	40 min	Upload lookups, definitions (incl. CIDR), automatic lookup, inputlookup, outputlookup
11	Create and schedule a report	8	35 min	Two reports, formatting, a schedule with a verifiable action, export, permissions
12	Build an interactive operational dashboard	9	50 min	Dashboard Studio: KPIs, charts, inputs, tokens, drilldown, live data
13	Configure a threshold-based alert	10	45 min	Checkout and brute-force alerts, throttling, webhook, triggering, troubleshooting
14	Troubleshoot a basic data-ingestion issue	11	40 min	Diagnose and fix a broken feed with <code>_internal</code> , preview, <code>btool</code> , conf files, fishbucket
15	Administration essentials	11	30 min	Directories, retention, roles and users, the Monitoring Console, admin searches
—	Challenge exercises	—	optional	Stretch tasks

Set up for Day 2

Throughout this guide, `$SPLUNK_HOME` means your Splunk installation directory. On the NovaMart lab servers this is `/home/ubuntu/splunk`. Set it once in every terminal you open:

```
export SPLUNK_HOME=/home/ubuntu/splunk # adjust if your Splunk is elsewhere (e.g. /opt/splunk)
$SPLUNK_HOME/bin/splunk version
```

Option A: you use the novamart_lab app (recommended)

- 1 Apps → Manage Apps → Install app from file. Choose `novamart_lab.tgz` (v1.1), tick **Upgrade app**, click **Upload**, then **Restart Splunk**.
- 2 Check Apps → Manage Apps: **NovaMart Lab** shows version **1.1.0**. Your existing data is not re-indexed.

The Day 2 files are now in `$SPLUNK_HOME/etc/apps/novamart_lab/data/` (`live/`, `tools/`, `troubleshoot/`), and the live-data inputs are already configured.

Option B: you onboarded the data manually on Day 1

Do not install the app (it would index everything a second time). Instead, unzip the v1.1 dataset over your Day 1 folder:

```
cd /home/ubuntu && unzip -o novamart_dataset.zip      # adds live/, tools/, troubleshoot/ and 4 new lookup
s
ls novamart_dataset
```

Wherever this guide writes `$SPLUNK_HOME/etc/apps/novamart_lab/data/`, use `/home/ubuntu/novamart_dataset/`. Lab 12 Part A shows the extra step to monitor the live files.

Check your data

```
| tstats count where index=novamart earliest=0 latest="09/21/2026:00:00:00" by sourcetype
```

EXPECTED

```
access_combined 22,641 · linux_secure 1,082 · novamart:app 3,042 (+ your Day 1 test lines) ·
novamart:payment 806 · novamart:firewall 300 if you did the optional UDP lab.
```

Conventions and time ranges

- Historical tasks (Labs 8–11) use **All time**. Once the live generator is running (from Lab 12), live events are added to `index=novamart`. To reproduce the historical counts exactly, use *Date Range: Between 14/09/2026 and 20/09/2026*, or add `earliest=0 latest="09/21/2026:00:00:00"` to the search.
- Your user time zone must be **Asia/Kolkata** (Day 1, Lab 0).
- Terminal commands run **on the Splunk server** (e.g. over SSH as `ubuntu`). Labs 12–14 need two or three terminals at once.
- Box labels: **EXPECTED** results · **CHECKPOINT** · **OPTIONAL** .

Build Statistical Searches Using stats and timechart

Goal: summarise NovaMart data with stats, chart and timechart; rank with top and rare; filter with where; calculate rates with eval.

Time range: **All time** (before the live generator is started in Lab 12).

Task 8.1 — Several functions in one stats

```
index=novamart sourcetype=access_combined
| stats count AS requests, dc(clientip) AS visitors, sum(bytes) AS bytes_sent,
    avg(bytes) AS avg_bytes, min(bytes) AS min_bytes, max(bytes) AS max_bytes by host
| eval avg_bytes=round(avg_bytes, 1)
```

EXPECTED

host	requests	visitors	bytes_sent	avg_bytes	min_bytes	max_bytes
web01	10856	414	52683824	4853.0	17	9500
web02	11785	417	58424723	4957.5	17	9500

Now remove `by host`. Why is `visitors` 425 and not 414 + 417?

Task 8.2 — count, count(field) and dc

```
index=novamart sourcetype=novamart:app
| stats count AS events, count(error_code) AS with_error_code, count(customer_id) AS with_customer, dc(customer_id) AS customers
```

EXPECTED

events 3,042 (+ test lines) · with_error_code 80 · with_customer 2,731 · customers 563

Task 8.3 — Group by region

```
index=novamart sourcetype=novamart:app action=checkout status=SUCCESS
| stats count AS orders, sum(amount) AS revenue, avg(amount) AS aov, max(amount) AS biggest, dc(customer_id) AS customers by region
| eval aov=round(aov, 0)
| sort - revenue
```

EXPECTED

region	orders	revenue	aov	biggest	customers
North	254	1770840	6972	56997	143
South	294	1717583	5842	56997	160
West	164	996654	6077	37998	99
East	100	707421	7074	56997	60

Task 8.4 — chart: two dimensions

```
index=novamart sourcetype=access_combined | chart count over host by status
```

Switch to the **Visualization** tab, choose **Column**, then *Format* → *Y-Axis* → *Scale: Log*. Why is the log scale useful here?

EXPECTED

2 rows. web01: 200=10,573 · 401=13 · 402=21 · 403=14 · 404=192 · 500=23 · 503=20. web02: 200=11,683 · 402=24 · 404=48 · 500=15 · 503=15, with empty cells for 401 and 403.

```
index=novamart sourcetype=novamart:app action=checkout status=SUCCESS
| chart sum(amount) over region by payment_method
```

EXPECTED

4 rows × 5 method columns. Check one cell: North / UPI = 702,275.

Task 8.5 — timechart

#	Search	Expected
a	<code>index=novamart sourcetype=access_combined timechart span=1d count by host</code>	7 rows; 18-Sep: web01 4,177, web02 4,480
b	<code>index=novamart sourcetype=access_combined earliest="09/18/2026:00:00:00" latest="09/19/2026:00:00:00" timechart span=1h count</code>	24 rows; 18:00 = 154, 19:00 = 1,731 , 20:00 = 1,722, 21:00 = 1,841 , 22:00 = 1,359
c	<code>index=novamart sourcetype=novamart:app action=checkout status=SUCCESS timechart span=1d sum(amount) AS revenue</code>	18-Sep: 3,092,024; other days 284,816 – 391,166
d	<code>index=novamart sourcetype=novamart:payment timechart span=1d avg(latency_ms) by gateway</code>	Normal days ~270–330 ms for both; 18-Sep: gw-alpha 494.2, gw-beta 4,731.6

For 8.5b, choose **Line** on the Visualization tab. For 8.5c, choose **Column**.

Task 8.6 — Time-based analysis with strftime and bin

```
index=novamart sourcetype=access_combined
| eval hour=strftime(_time, "%H")
| stats count by hour
| sort - count
```

EXPECTED

Busiest: 21 (3,039), 20 (2,924), 19 (2,847). Quietest: 04 (63), 02 (68), 01 (101).

```
index=novamart sourcetype=access_combined
| eval weekday=strftime(_time, "%A")
| stats count by weekday | sort - count
```

EXPECTED

Friday 8,657 · Sunday 2,541 · Saturday 2,501 · Wednesday 2,333 · Tuesday 2,216 · Thursday 2,197 · Monday 2,196

```
index=novamart sourcetype=access_combined uri_path=/checkout
| bin _time span=1d
| stats count AS checkouts, count(eval(status>=500)) AS errors by _time
```

EXPECTED

7 rows; errors = 0 on every day except 18-Sep (480 checkouts, 35 errors).

Task 8.7 — top and rare

#	Search	Expected
a	<code>index=novamart sourcetype=novamart:app action=checkout status=SUCCESS top limit=2 product_id by region</code>	8 rows. North: NM-1002 (33), NM-1001 (31) · South: NM-1002 and NM-1003 (37 each) · East: NM-1001 (13), NM-1002 (10) · West: NM-1002 (25), NM-1001 (24)
b	<code>index=novamart sourcetype=novamart:app action=checkout rare limit=3 payment_method</code>	WALLET 53 · COD 86 · NETBANKING 98
c	<code>index=novamart sourcetype=access_combined top limit=3 clientip showperc=false countfield=requests</code>	198.51.100.23 715 · 10.20.0.5 336 · 192.0.2.66 180

Task 8.8 — where

#	Search	Expected
a	<code>index=novamart sourcetype=access_combined stats count by clientip where count > 300</code>	2 rows: 198.51.100.23 (715), 10.20.0.5 (336)
b	<code>index=novamart sourcetype=access_combined stats count by useragent where like(useragent, "%bot%")</code>	1 row: Googlebot (39)

#	Search	Expected
c	<code>index=novamart sourcetype=novamart:app action=checkout status=SUCCESS stats sum(amount) AS revenue by region where region="East" OR revenue > 1700000</code>	3 rows: East, North, South
d	Same as c, but with <code>where region=East</code> (no quotes)	Only North and South. Why?

Task 8.9 — eval with stats: error rate per host

```
index=novamart sourcetype=access_combined
| stats count AS total, count(eval(status ≥ 500)) AS server_errors, count(eval(status ≥ 400 AND status < 500)) AS client_errors by host
| eval server_err_pct=round(server_errors/total*100, 3), client_err_pct=round(client_errors/total*100, 2)
```

EXPECTED

web01: 10,856 total, 43 server (0.396%), 240 client (2.21%). web02: 11,785 total, 30 server (0.255%), 72 client (0.61%). web01's client errors are high because of the scanner.

CHECKPOINT

- ☐ You can produce the same data as a long table (stats), a matrix (chart) and a trend (timechart).
- ☐ You can filter aggregated results with `where` and quote strings correctly.
- ☐ You can calculate a percentage using `count(eval(...))` and `eval`.

Analyse Application Performance and Error Trends

Goal: act as NovaMart's SRE. Measure response time and gateway latency, trend errors by day and service, and write up your findings.

Task 9.1 — Checkout response time by payment method

```
index=novamart sourcetype=novamart:app action=checkout
| stats count, avg(response_ms) AS avg_ms, max(response_ms) AS max_ms by payment_method
| eval avg_ms=round(avg_ms, 0)
| sort - avg_ms
```

EXPECTED

payment_method	count	avg_ms	max_ms
CARD	336	3204	30883
NETBANKING	98	2338	30539
WALLET	53	413	844
UPI	319	392	900
COD	86	345	833

Now add `status=SUCCESS` to the first line and run it again. CARD drops to ≈ 651 ms and NETBANKING to ≈ 518 ms. What does that tell you about *averages* when a few requests take 30 seconds?

Task 9.2 — Gateway latency: the week and the incident

```
index=novamart sourcetype=novamart:payment
| stats count, avg(latency_ms) AS avg_ms, median(latency_ms) AS median_ms, perc95(latency_ms) AS p95_ms, max(latency_ms) AS max_ms by gateway
| eval avg_ms=round(avg_ms, 0)
```

EXPECTED

gw-alpha: 372 payments, avg ≈ 395 , median ≈ 356 , p95 ≈ 812 , max 900 ms. gw-beta: 434 payments, avg $\approx 3,008$, median ≈ 411 , p95 $\approx 30,260$, max 30,883 ms. (Percentiles may differ slightly from these values because of how Splunk estimates them.) The medians are similar, so gw-beta's typical payment is healthy; its *tail* is where the problem is.

Zoom into the incident at 5-minute resolution:

```
index=novamart sourcetype=novamart:payment earliest="09/18/2026:20:00:00" latest="09/18/2026:21:00:00"
| timechart span=5m avg(latency_ms) by gateway
```

EXPECTED

gw-beta: 20:00 ≈ 646 → 20:05 ≈ 30,375 · stays at 26,000–30,400 until 20:40 · 20:45 ≈ 5,515 (failover)
· 20:50 ≈ 455. gw-alpha stays at 240–760 ms throughout. Display it as a **Line** chart.

Task 9.3 — Error and warning trend by day

```
index=novamart sourcetype=novamart:app
| rex "^S+s(?:<log_level>[A-Z]+)\s+[(?:<service>[^\]]+)]\"
| search log_level IN (ERROR, WARN)
| timechart span=1d count by log_level
```

EXPECTED

_time	14	15	16	17	18	19	20
ERROR	2	6	4	9	74	5	8
WARN	8	7	19	19	42	14	10

Which service produces the ERRORS on the *quiet* days? Replace the last line with `| search log_level=ERROR | chart count over service by date_mday` (or `timechart span=1d count by service`).

EXPECTED

`search-service` logs errors **every day** (2, 6, 4, 9, 4, 5, 8 = 38 in total): a chronic problem hidden by the big outage. `order-service` and `payment-service` errors occur only on 18-Sep (35 each).

Task 9.4 — Web error trend

```
index=novamart sourcetype=access_combined status>=400
| eval class=if(status>=500, "5xx", "4xx")
| timechart span=1d count by class
```

EXPECTED

4xx: 8, 13, **191**, 14, 56, 15, 15 · 5xx: 2, 6, 4, 9, **39**, 5, 8. The 4xx spike on 16-Sep is the vulnerability scanner; the 5xx spike on 18-Sep is the outage (35 × 503) plus 4 search errors.

Task 9.5 — Checkout success rate per day

```
index=novamart sourcetype=novamart:app action=checkout
| bin _time span=1d
| stats count AS attempts, count(eval(status="SUCCESS")) AS ok, count(eval(status="FAILED")) AS failed, count(eval(status="DECLINED")) AS declined by _time
| eval success_pct=round(ok/attempts*100, 1)
```

EXPECTED

14-Sep 95.8% · 15-Sep 98.3% · 16-Sep 92.4% · 17-Sep 92.9% · **18-Sep 87.7%** (480 attempts: 421 ok, 35 failed, 24 declined) · 19-Sep 94.6% · 20-Sep 95.8%

Task 9.6 — Write up your findings

Using only your results from this lab, complete the SRE summary. Model answers are in Appendix A.

SRE WEEKLY SUMMARY

1. Biggest incident and its measurable impact:

2. Which dependency, and what the latency data shows:

3. A chronic issue that needs a ticket:

4. A security-related observation from the error trends:

5. One metric you would put on a dashboard, and why:

CHECKPOINT

- ☐ You can explain why p95/median complement averages.
- ☐ You found the chronic search-service errors.
- ☐ You can produce a per-day success rate with `bin` + `stats` + `eval`.

Enrich Data Using CSV Lookups

Goal: upload the Day 2 lookup tables, create definitions (including CIDR matching), enrich searches manually and automatically, filter with inputlookup, and create a table with outputlookup.

Task 10.1 — Upload the lookup files

The four Day 2 CSV files are in the dataset's `lookups/` folder on your laptop.

- 1 *Settings* → *Lookups* → *Lookup table files* → *New Lookup Table File*. Destination app: **search**. Upload `network_zones.csv` with destination file name `network_zones.csv`. Click **Save**.
- 2 Repeat for `region_targets.csv`, `gateway_sla.csv` and `known_bots.csv`.
- 3 For each of the four files, click **Permissions** → *This app only (search)* → Read: **Everyone**, Write: admin → **Save**.

```
| inputlookup region_targets.csv
| inputlookup known_bots.csv where policy="block"
```

EXPECTED

4 rows (North, South, East, West with managers and targets); 2 blocked bots (198.51.100.23 Content scraper, 192.0.2.66 Nikto vulnerability scanner).

Task 10.2 — Create lookup definitions

- 1 *Settings* → *Lookups* → *Lookup definitions* → *New Lookup Definition*. Destination app **search**, Name `region_targets`, Type *File-based*, Lookup file `region_targets.csv`. **Save**. Repeat for `gateway_sla` and `known_bots`.
- 2 Create `network_zones` the same way, but tick **Advanced options** and set **Match type** = `CIDR(cidr)`. **Save**.
- 3 Share all four definitions: **Permissions** → *This app only*, Read = Everyone.

Option A (app) users already have `novamart_products`, `http_status` and `threat_intel_ips` from the app; Option B users created them in Day 1 Lab 6.

TEST THE CIDR DEFINITION

```
| makeresults | eval src_ip="10.20.1.15" | lookup network_zones cidr AS src_ip OUTPUT zone, owner
```

EXPECTED

`zone = Admin workstations (VPN)`, `owner = IT Operations`. If both are empty, the Match type was not set: edit the definition.

Task 10.3 — Revenue against target

```
index=novamart sourcetype=novamart:app action=checkout status=SUCCESS
| stats sum(amount) AS revenue by region
| lookup region_targets region OUTPUT region_manager, weekly_revenue_target_inr AS target
| eval attainment_pct = round(revenue / target * 100, 1),
      status = if(attainment_pct ≥ 100, "On target", "Below target")
| sort - attainment_pct
```

EXPECTED

North 118.1% (Aditi Rao) · South 107.3% (Karthik Iyer) · West 90.6% (Meera Kulkarni) · East 88.4% (Sourav Das)

Task 10.4 — Network zones for SSH logins (CIDR)

```
index=novamart sourcetype=linux_secure "Accepted"
| rex "Accepted (?<auth_method>\S+) for (?<user>\S+) from (?<src_ip>\S+)"
| lookup network_zones cidr AS src_ip OUTPUT zone, trust
| fillnull value="Internet" zone
| fillnull value="external" trust
| stats count, values(user) AS users, values(auth_method) AS methods by zone, trust
```

EXPECTED

Admin workstations (VPN) / internal: 76, users ananya deploy devops rahul, publickey. **Internet / external: 1, deploy, password.**

Now do the same for **failed** passwords (replace the rex with "Failed password for (invalid user)?(?<user>\S+) from (?<src_ip>\S+)").

EXPECTED

Internet 452 · Admin workstations (VPN) 14 (admins mistyping their passwords).

Task 10.5 — Latency against SLA

```
index=novamart sourcetype=novamart:payment
| stats perc95(latency_ms) AS p95_ms, avg(latency_ms) AS avg_ms by gateway
| lookup gateway_sla gateway OUTPUT provider, sla_p95_latency_ms AS sla_ms, support_contact
| eval avg_ms=round(avg_ms, 0), breach=if(p95_ms > sla_ms, "YES", "no")
```

EXPECTED

gw-alpha (Payment Partner A): p95 ≈ 812 vs SLA 800 → **YES**, a small breach. gw-beta (Payment Partner B): p95 ≈ 30,260 vs 1,000 → **YES**. Run it again for 19–20 Sep only: both gateways are within SLA.

Task 10.6 — Filter out bots with inputlookup (subsearch)

```
index=novamart sourcetype=access_combined NOT [| inputlookup known_bots | fields clientip]
| stats count AS human_requests
```

```
index=novamart sourcetype=access_combined
| lookup known_bots clientip OUTPUT bot_name, policy
| fillnull value="Human" bot_name
| stats count by bot_name, policy
| eventstats sum(count) AS total
| eval pct=round(count/total*100, 2)
```

EXPECTED

Human requests: **21,371**. Bots: Content scraper 715 (block) · Load balancer health check 336 (allow) · Nikto vulnerability scanner 180 (block) · Googlebot 39 (allow). Human = 94.39%.

THINK ABOUT IT

What does the subsearch `[| inputlookup known_bots | fields clientip]` turn into before the main search runs? Click **Job** → **Inspect Job** and look at `normalizedSearch`.

Task 10.7 — Automatic lookup

- 1 *Settings* → *Lookups* → *Automatic lookups* → *New Automatic Lookup*. Destination app **search**. Name `nm_products_auto`. Lookup table `novamart_products`. Apply to **sourcetype** named `novamart:app`. Lookup input fields: `product_id` = `product_id`. Lookup output fields: `product_name` = `product_name`, `category` = `category`. **Save**, then set Permissions to *This app only*, Read = Everyone.

- 2 Test it with no `lookup` command:

```
index=novamart sourcetype=novamart:app action=checkout status=SUCCESS | top category
index=novamart sourcetype=novamart:app action=checkout status=SUCCESS category=Electronics | stats
count, sum(amount) by region
```

EXPECTED

Orders by category: Electronics 389 · Fashion 117 · Home & Kitchen 101 · Beauty 92 · Sports 67 · Books 46. Electronics revenue: North 1,450,548 · South 1,357,136 · West 838,197 · East 581,340. `product_name` and `category` now appear in the fields sidebar for every `novamart:app` event that has a `product_id`.

Task 10.8 — Build a table with outputlookup, then use it

```
index=novamart sourcetype=linux_secure "Accepted publickey"
| rex "From (?<src_ip>\S+)"
| stats count AS logins, max(_time) AS last_seen by src_ip
| eval last_seen=strftime(last_seen, "%Y-%m-%d %H:%M")
| outputlookup nm_admin_ips.csv
```

EXPECTED

4 rows written: 10.20.1.15 (23), 10.20.1.22 (22), 10.20.1.31 (12), 10.20.1.44 (19). Check them with `| inputlookup nm_admin_ips.csv`.

LOGINS FROM ANYWHERE NOT ON THE ADMIN LIST

```
index=novamart sourcetype=linux_secure "Accepted"
| rex "Accepted (?<auth_method>\$+) for (?<user>\$+) from (?<src_ip>\$+)"
| search NOT [| inputlookup nm_admin_ips.csv | fields src_ip]
| table _time, user, auth_method, src_ip
```

EXPECTED

Exactly one event: 17-Sep 02:34:51, deploy, password, 203.0.113.77.

CHECKPOINT

- ☐ Four new lookup files and definitions exist and are shared.
- ☐ The CIDR lookup resolves 10.20.1.15.
- ☐ `category` appears automatically on `novamart:app` events.
- ☐ `nm_admin_ips.csv` exists and isolates the attacker's login.

Create and Schedule a Report

Goal: build a chart report and a table report for NovaMart leadership, schedule one with an action you can verify, export results, and set permissions.

Task 11.1 — Report 1: daily revenue by category (chart)

```
index=novamart sourcetype=novamart:app action=checkout status=SUCCESS earliest="09/14/2026:00:00:00" latest="09/21/2026:00:00:00"
| lookup novamart_products product_id OUTPUT category
| timechart span=1d sum(amount) by category
```

- 1 **Visualization** tab → **Column. Format** → **General** → **Stack Mode: Stacked. Format** → **Y-Axis** → **Title: Revenue (INR). Format** → **Legend** → **Position: Bottom.**
- 2 **Save As** → **Report.** Title **NM - Report - Daily Revenue by Category**. Description **Successful order revenue per day and category (INR).** Content: **Column Chart.** Time Range Picker: **No.** **Save**, then **View.**

EXPECTED

7 stacked columns. 18-Sep towers at ₹30,92,024, of which Electronics is ₹26,84,885. Hover over segments to see the values. The inline `earliest / latest` keep the report on the lab week even after live data arrives.

Task 11.2 — Report 2: payment gateway performance (formatted table)

```
index=novamart sourcetype=novamart:payment earliest="09/14/2026:00:00:00" latest="09/21/2026:00:00:00"
| stats count AS payments, count(eval(result="approved")) AS approved, count(eval(result="timeout")) AS timeouts,
    avg(latency_ms) AS avg_ms, max(latency_ms) AS max_ms by gateway
| eval approval_pct=round(approved/payments*100, 1), avg_ms=round(avg_ms, 0)
| lookup gateway_sla gateway OUTPUT provider, sla_p95_latency_ms AS sla_ms
| table gateway, provider, payments, approved, approval_pct, timeouts, avg_ms, max_ms, sla_ms
```

- 1 On the **Statistics** tab, click the paintbrush on **approval_pct** → **Color: Ranges** (red below 92, yellow 92–96, green above 96) → **Number format: 1 decimal, unit %**.
- 2 Paintbrush on **max_ms** → **Color: Ranges** (red above 5000). **Format** → **Data Overlay: None**, **Row numbers: Yes.**
- 3 **Save As** → **Report:** **NM - Report - Payment Gateway Performance**, Content: **Statistics Table.**

EXPECTED

gw-alpha: Payment Partner A, 372 payments, 350 approved, 94.1%, 0 timeouts, avg ≈ 395, max 900, SLA 800. gw-beta: Payment Partner B, 434, 376, 86.6%, 35 timeouts, avg ≈ 3,008, max 30,883, SLA 1000.

Task 11.3 — Schedule Report 1 with a verifiable action

Email needs a mail server, so we use **Output results to lookup**. That writes the results of every scheduled run to a CSV file you can check.

- 1 **Reports** → find `NM - Report - Daily Revenue by Category` → **Edit** → **Edit Schedule**. Tick **Schedule Report**.
- 2 Schedule: **Run on Cron Schedule**, Cron Expression `*/2 * * * *` (every 2 minutes, for testing). Time Range: leave as is (the SPL sets its own). Schedule Priority: Default. Schedule Window: No window.
- 3 **+ Add Actions** → **Output results to lookup**. Lookup name `nm_daily_revenue.csv`, **Replace** (not append). **Save**.
- 4 Wait up to 3 minutes, then check:

```
| inputlookup nm_daily_revenue.csv  
  
index=_internal sourcetype=scheduler savedsearch_name="NM - Report - Daily Revenue by Category"  
| table _time, status, run_time, result_count, alert_actions
```

EXPECTED

The lookup has 7 rows (one per day) with a column per category. `scheduler.log` shows `status=success`, `result_count=7`, `alert_actions=lookup` every 2 minutes.

- 5 Now make it production-ready: change the cron to `0 7 * * *` (daily at 07:00). In production you would also replace the fixed dates with `earliest=-1d@d latest=@d`.

Optional (email): if your trainer has configured *Settings* → *Server settings* → *Email settings*, add a **Send email** action with *Attach PDF* ticked and your address in **To**.

Task 11.4 — Export and share

- 1 Open Report 2. Click **Export** (the download icon) → Format **CSV** → file name `gateway_performance` → **Export**. Open the file on your laptop.
- 2 Open Report 1 → **Export** → **Export PDF** (or **Print** → **Save as PDF**).
- 3 **Reports** → **Edit** → **Edit Permissions** on Report 2: Display For **App**, Run As **Owner**, Read: **Everyone**, Write: **admin**. **Save**. Leave Report 1 *Private* for now; you will check both in Lab 15 as a restricted user.

THINK ABOUT IT

Report 2 shows only payment data. Would *Run As: Owner* be acceptable for a report on `sourcetype=linux_secure` shared with Everyone? Why not?

Task 11.5 — Add a report to a dashboard

Open Report 2 → **Add to Dashboard** → **New** → Title `NM Management Pack`, Permissions *Shared in App*, Dashboard type **Classic** → Panel powered by **Report**, Display as **Statistics** → **Save** → **View Dashboard**. Add Report 1 to the same dashboard as a Column Chart.

CHECKPOINT

- ☐ Two reports exist with the `NM - Report -` prefix.
- ☐ The schedule ran (`scheduler.log` shows success) and `nm_daily_revenue.csv` was written.
- ☐ You exported CSV and PDF, and set permissions on Report 2.

Build an Interactive Operational Dashboard

Goal: start live NovaMart traffic, then build the "NovaMart Operations" dashboard in Dashboard Studio, with KPI single values, charts, inputs, tokens and a drilldown.

Part A — Start live traffic

Open a terminal on the Splunk server (**Terminal 1**). Leave the generator running for the rest of the day.

OPTION A: NOVAMART_LAB APP

```
export SPLUNK_HOME=/home/ubuntu/splunk
cd $SPLUNK_HOME/etc/apps/novamart_lab/data/live
nohup python3 novamart_live.py --mode normal --minutes 480 > /tmp/nm_live_normal.out 2>&1 &
sleep 35; cat /tmp/nm_live_normal.out
```

OPTION B: MANUAL DAY 1 SETUP (ALSO ADD THREE MONITOR INPUTS)

```
export SPLUNK_HOME=/home/ubuntu/splunk
cd /home/ubuntu/novamart_dataset/live
nohup python3 novamart_live.py --mode normal --minutes 480 > /tmp/nm_live_normal.out 2>&1 &
sleep 5
S=$SPLUNK_HOME/bin/splunk; L=/home/ubuntu/novamart_dataset/live; A="-auth admin:<your-password>"
$$ add monitor $L/web_live_access.log -index novamart -sourcetype access_combined -hostname web03 $A
$$ add monitor $L/app_live.log -index novamart -sourcetype novamart:app -hostname app01 $A
$$ add monitor $L/secure_live.log -index novamart -sourcetype linux_secure -hostname bastion01 $A
```

```
index=novamart earliest=-5m | stats count by host, sourcetype
```

EXPECTED

The generator prints "Writing normal traffic ... for 480 min" and then a progress line every 30 seconds. After a minute, the search shows `web03 / access_combined` (about 100–150 events a minute), `app01 / novamart:app` (a few checkouts a minute), and occasionally `bastion01 / linux_secure`. All of them have the **current** time.

Part B — Create the dashboard and inputs

- 1 **Search & Reporting** → **Dashboards** → **Create New Dashboard**. Title `NovaMart Operations`, Description *Platform health, payments and business KPIs*, Permissions **Shared in App**. Choose **Dashboard Studio** and layout **Grid**. **Create**.
- 2 **Time range input**: Studio dashboards normally start with a global time range input (token `global_time`). If yours has none, use **Add Input** → **Time Range** and name the token `global_time`. Set its default to *Date Range: between 14/09/2026 and 20/09/2026*.
- 3 **Add Input** → **Dropdown**. Title `Host`, Token name `host`. Static menu items (label / value): `All` / `*`, `web01` / `web01`, `web02` / `web02`, `web03 (live)` / `web03`. Default selected value: `*`.
- 4 **Add Input** → **Dropdown**. Title `Region`, Token name `region`. Items: `All` / `*`, `North`, `South`, `East`, `West`. Default `*`.

Menu names differ slightly between Splunk versions (e.g. *Add input* vs the input icon in the toolbar). The concepts and token names are what matter.

Part C — KPI single values (top row)

For each panel: **Add Chart** → **Single Value** → in the data source area click **Create search** (or + *Setup primary data source* → *Search*), give the data source a name, paste the SPL, and apply. Set the panel **Title**.

Title	SPL	Options
Requests	<code>index=novamart sourcetype=access_combined host=\$host\$ stats count</code>	Number format with thousands separator
Server errors (5xx)	<code>index=novamart sourcetype=access_combined host=\$host\$ status ≥ 500 stats count</code>	<i>Coloring</i> → <i>Dynamic</i> : 0–4 green, 5–19 amber, 20+ red
Successful orders	<code>index=novamart sourcetype=novamart:app action=checkout status=SUCCESS region=\$region\$ stats count</code>	—
Revenue (INR)	<code>index=novamart sourcetype=novamart:app action=checkout status=SUCCESS region=\$region\$ stats sum(amount)</code>	Unit ₹ (before), thousands separator

EXPECTED (TIME 14–20 SEP, HOST ALL, REGION ALL)

Requests **22,641** · Server errors **73** (red) · Successful orders **812** · Revenue **₹5,192,498**.
Host = web01 → 10,856 and 43 · Host = web02 → 11,785 and 30. Region = East → 100 orders, ₹707,421. The host token does not affect the order panels, and the region token does not affect the web panels: tokens only change searches that use them.

Part D — Charts

Title / type	SPL
Requests over time <i>Line</i>	<code>index=novamart sourcetype=access_combined host=\$host\$ timechart count by host</code>
Server errors over time <i>Column (stacked)</i>	<code>index=novamart sourcetype=access_combined host=\$host\$ status ≥ 500 timechart count by status</code>
Revenue by category <i>Bar</i>	<code>index=novamart sourcetype=novamart:app action=checkout status=SUCCESS region=\$region\$ lookup novamart_products product_id OUTPUT category stats sum(amount) AS revenue by category sort - revenue</code>
Payment latency by gateway <i>Line</i>	<code>index=novamart sourcetype=novamart:payment timechart avg(latency_ms) by gateway</code>

EXPECTED

For the lab week, the two line charts show the Friday spike. The error columns peak on 18-Sep (39: 35 × 503 and 4 × 500). Revenue by category: Electronics 4,227,221 at the top. Payment latency: gw-beta jumps on 18-Sep. Resize panels by dragging their edges; drag them to arrange rows.

Part E — Drilldown: click a status to see its events

- 1 Add a **Table** titled `Errors by status` with `index=novamart sourcetype=access_combined host=$host$ status ≥ 400 | stats count by status | sort - count`.

2 Select the table → configuration panel → **Interactions** → + **Add Interaction** → On click: **Set tokens** → Token name `sel_status`, Token value: the **status** column of the clicked row (`row.status.value`). Apply.

3 Add a second **Table** titled `Events for selected status` with:

```
index=novamart sourcetype=access_combined host=$host$ status=$sel_status$
| table _time, host, clientip, method, uri_path, status, useragent
| head 100
```

4 **Save and View.** The detail table waits for input until you click a row in *Errors by status*. Click `503`.

EXPECTED

Errors by status (All hosts): 404 240 · 402 45 · 500 38 · 503 35 · 403 14 · 401 13. Clicking **503** fills the detail table with 35 rows, all POST `/checkout` on 18-Sep between 20:07 and 20:44. Change Host to web02 and click 503 again: 15 rows.

Part F — A live panel

Add a **Line** chart titled `Live: requests per minute (web03)` with its own time window, set inline so that it ignores the global picker:

```
index=novamart sourcetype=access_combined host=web03 earliest=-15m latest=now | timechart span=1m count
```

In the dashboard's view mode, set an automatic refresh (e.g. the data source's *refresh* option to `1m`, or reload the page). The line keeps moving while the generator runs.

EXPECTED

About 100–150 requests per minute, with a new point every minute.

Part G — Arrange, save and share

- ☐ Top row: the four KPIs. Second row: requests over time, server errors over time. Third row: errors by status + events for selected status. Fourth row: revenue by category, payment latency, live panel.
- ☐ **Save.** Look at **Source** (the JSON definition) and find your `$host$` token inside a query.
- ☐ Download it as **PDF** or **PNG** (Export/Download menu).
- ☐ Check the permissions: Shared in App, Read = Everyone.

STUCK, OR PREFER CLASSIC?

Appendix C contains the complete dashboard as Classic Simple XML. Create a new *Classic* dashboard, open *Edit* → *Source*, replace everything with the XML, and save.

CHECKPOINT

- ☐ The live generator is running, and `host=web03` has current events.
- ☐ The dashboard has 4 KPIs, 4 charts, a drilldown table pair and a live panel.
- ☐ Changing Host or Region changes only the panels that use that token.

Configure a Threshold-Based Alert

Goal: backtest, create and trigger a checkout-error alert with throttling and a webhook; build a per-result brute-force alert; troubleshoot both with `_internal`.

You need the live generator from Lab 12 running (normal mode) and two extra terminals.

Part A — Start the webhook receiver (Terminal 2)

```
export SPLUNK_HOME=/home/ubuntu/splunk
python3 $SPLUNK_HOME/etc/apps/novamart_lab/data/tools/webhook_listener.py
# Option B: python3 /home/ubuntu/novamart_dataset/tools/webhook_listener.py
```

EXPECTED

NovaMart webhook listener on `http://0.0.0.0:8765` Leave this terminal visible: alert payloads will appear here.

Part B — Backtest the alert logic on last week's data

```
index=novamart sourcetype=access_combined uri_path=/checkout status>=500
earliest="09/18/2026:19:00:00" latest="09/19/2026:00:00:00"
| bin _time span=5m
| stats count AS errors by _time
| eval would_fire=if(errors>=5, "YES", "no")
```

EXPECTED

8 windows: 20:05 → 3 · 20:10 → 3 · **20:15** → **5 · 20:20** → 7 · 20:25 → 2 · **20:30** → 7 · 20:35 → 3 · **20:40** → 5. Four windows reach the threshold; the first fires at about 20:20. Run it over 14–17 and 19–20 Sep: **no** results, so there are no false alarms.

Part C — Build the live alert search

Set the time picker to **Last 15 minutes** and run:

```
index=novamart sourcetype=access_combined uri_path=/checkout status>=500 earliest=-5m@m latest=@m
| stats count AS errors, dc(clientip) AS customers, min(_time) AS first, max(_time) AS last
| eval window_start=strftime(first, "%H:%M:%S"), window_end=strftime(last, "%H:%M:%S")
| fields errors, customers, window_start, window_end
| where errors >= 5
```

EXPECTED

No results: normal traffic has no checkout 5xx. (Remove the last line and you get one row with `errors=0`.)

Part D — Save it as a throttled alert with actions

- 1 **Save As → Alert.** Title `NM - Alert - Checkout 5xx spike`. Description *5+ HTTP 5xx on /checkout in 5 minutes. Runbook: check payment gateways and the circuit breaker.* Permissions: **Shared in App.**
- 2 Alert type: **Scheduled** → **Run on Cron Schedule** → Cron `* * * * *`. (Every minute for the lab. It overlaps the 5-minute window on purpose, making a rolling window; throttling prevents duplicates. In production you would use `*/5` with `-5m@m / @m`.)
- 3 Trigger alert when: **Number of Results** is greater than `0`. Trigger: **Once**. Tick **Throttle** → Suppress triggering for `10` minutes.
- 4 + **Add Actions** → **Add to Triggered Alerts**, Severity **High**.
- 5 + **Add Actions** → **Webhook**, URL `http://127.0.0.1:8765/checkout-alert`. **Save.**

If your trainer has configured email, also add **Send email** with subject `[SEV-2] $name$ - $result.errors$ checkout errors`.

Part E — Trigger it (Terminal 3)

```
export SPLUNK_HOME=/home/ubuntu/splunk
cd $SPLUNK_HOME/etc/apps/novamart_lab/data/live # Option B: cd /home/ubuntu/novamart_dataset/live
python3 novamart_live.py --mode incident --minutes 6
```

The incident mode writes a failed checkout (HTTP 503, plus gw-beta timeout errors in the app log) about every 7 seconds, alongside the normal traffic from Terminal 1.

EXPECTED, WITHIN ABOUT 2 MINUTES

- **Terminal 2** prints a POST to `/checkout-alert`, with `"search_name": "NM - Alert - Checkout 5xx spike"` and a `result` such as `{"errors": "8", "customers": "8", "window_start": "...", "window_end": "..."}.`
- *Activity* → *Triggered Alerts* shows the alert with severity High.
- Only **one** notification arrives for the next 10 minutes, even though the condition stays true: that is throttling at work.

While the incident runs, open your **NovaMart Operations** dashboard with the time range *Last 15 minutes* and Host = web03. The Server errors KPI turns red.

Part F — A per-result brute-force alert

```
index=novamart sourcetype=linux_secure "Failed password" earliest=-5m@m latest=@m
| rex "Failed password for (invalid user )?(?<user>\$+) from (?<src_ip>\d{1,3}(\.?\d{1,3}){3})"
| stats count AS failures, dc(user) AS users, values(user) AS tried by src_ip
| where failures > 20
```

- 1 **Save As → Alert:** `NM - Alert - SSH brute force`, Scheduled, cron `* * * * *`.
- 2 Trigger: Number of Results > 0, **For each result**. Throttle: tick, **Suppress results containing field value** `src_ip`, for `60` minutes.
- 3 Actions: **Add to Triggered Alerts** (Critical) and **Webhook** `http://127.0.0.1:8765/bruteforce-alert`. **Save.**

4 In Terminal 3: `python3 novamart_live.py --mode bruteforce --minutes 3`

EXPECTED

Within about 2 minutes, one webhook POST to `/bruteforce-alert` with `"src_ip": "203.0.113.99"`, `failures` above 20, and `users` up to 8. There is no second notification for that IP within 60 minutes.

Backtest it: replace `earliest=-5m@m latest=@m` with `earliest="09/17/2026:00:00:00" latest="09/18/2026:00:00:00"`, and add `| bin _time span=5m` before `stats` and `_time` to its `by` clause.

EXPECTED (BACKTEST)

Five rows, all 203.0.113.77, 17-Sep: 02:10 → 73, 02:15 → 83, 02:20 → 87, 02:25 → 88, 02:30 → 83 failures, with 11–12 different user names in each window. The first alert would have fired about 20 minutes before the attacker's successful login at 02:34:51.

Part G — Troubleshoot and manage

```
index=_internal sourcetype=scheduler savedsearch_name="NM - Alert - *"
| table _time, savedsearch_name, status, result_count, alert_actions, suppressed
| sort - _time
```

```
index=_internal sourcetype=splunkd component=sendmodalert
| table _time, action, log_level, event_message | sort - _time
```

EXPECTED

One scheduler line per alert per minute. `result_count=0` on quiet minutes. When the condition is met: `result_count ≥ 1` and `alert_actions` listing the actions that ran (e.g. `webhook`), with `suppressed=1` during throttling. The `sendmodalert` lines show the webhook invocations, with any error if the listener was not running.

BREAK IT ON PURPOSE

Stop the listener in Terminal 2 (Ctrl+C), run `--mode incident --minutes 3` again after the 10-minute throttle has expired, and find the webhook error in `sendmodalert`.

IMPORTANT · CLEAN UP

When you are done, **disable both alerts** (*Alerts* → *Edit* → *Disable*): every-minute alerts should not run all day. Keep the normal traffic generator running for Lab 15.

CHECKPOINT

- ☐ You backtested both alerts and can say when they would have fired.
- ☐ The checkout alert fired once (throttled) and delivered a webhook.
- ☐ The brute-force alert fired once per source IP.
- ☐ You can find the run history and action results in `_internal`.

Troubleshoot a Basic Data-Ingestion Issue

Goal: deploy the warehouse vendor's input exactly as delivered, find out why no data appears (and what else is wrong), fix it with .conf files, re-index, and validate.

SCENARIO

NovaMart's Mumbai warehouse (WH-MUM) will send stock movements to Splunk. The vendor supplied the log file and an `inputs.conf`. Requirements from the supply-chain team: data in index `novamart_inv`, source type `novamart:inventory`, host `wh-mum01`, and correct event times (the `ts=` value, in IST).

A LINE FROM INVENTORY_WH.LOG

```
WH-MUM batch_date=2026-08-10 sku=NM-3002 event=LOW_STOCK qty=2 ts=14-09-2026 06:05:57.128 operator=svc_inventory
```

Part A — Deploy the vendor's configuration as delivered

```
export SPLUNK_HOME=/home/ubuntu/splunk
mkdir -p $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local
cp $SPLUNK_HOME/etc/apps/novamart_lab/data/troubleshoot/inputs_broken.conf \
  $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local/inputs.conf
cat $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local/inputs.conf
$SPLUNK_HOME/bin/splunk restart
```

Option B (no app): copy from `/home/ubuntu/novamart_dataset/troubleshoot/inputs_broken.conf` instead, then point the monitor at your file:

```
sed -i 's#\${SPLUNK_HOME}/etc/apps/novamart_lab/data/troubleshoot#/home/ubuntu/novamart_dataset/troubleshoot#' $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local/inputs.conf
```

Creating a small app (`nm_inventory_inputs`) for one data source is good practice: everything about the feed lives in one folder that you can version, move or remove.

Part B — Symptom: "there's no data"

- 1 Search `index=novamart_inv` over **All time**. Note the result, and any message under the search bar.
- 2 Open **Messages** in the Splunk bar. Is there a new message?
- 3 Ask splunkd:

```
index=_internal sourcetype=splunkd novamart_inv earliest=-30m
| table _time, log_level, component, event_message
```

EXPECTED

No events. Messages and `splunkd.log` show a warning like *"Received event for unconfigured/disabled/deleted index=novamart_inv with source='source::.../inventory_wh.log' host='host::<your-server-name>' sourcetype='sourcetype::novamart:inventory' ... So far received events from 1 missing index(es)."*

DIAGNOSE

Fault 1: What is wrong with the index?

Fault 2: Look closely at `host=` in the warning. Does it meet the requirement?

Part C — Check the timestamps before you index anything

Use the Add Data **preview** as a safe test bench: you will not submit anything.

- 1 *Settings* → *Add Data* → *Upload* → select `troubleshoot/inventory_wh.log` from the dataset on your laptop → **Next**.
- 2 Look at the `_time` column in the preview and compare it with the `ts=` value in each event.

EXPECTED

The dates are in **August 2026** (taken from `batch_date=2026-08-..`), not 14–20 September. Splunk uses the first date-like text it finds in the event. This is **Fault 3**.

- 3 Fix it in the preview: **Timestamp** → **Advanced...** → Timestamp format `%d-%m-%Y %H:%M:%S.%3N`, Timestamp prefix `ts=`, Lookahead `25`, Time Zone *(GMT+05:30) Asia/Kolkata* → **Apply settings**. The first event now shows **14/09/2026 06:05:57.128**.
- 4 Note the settings, then **cancel** the wizard (click the Splunk logo). Do **not** submit: the real input is the monitor from Part A.

Part D — Fix all three faults in configuration files

1. CREATE THE INDEX (IN THE SAME APP)

```
cat > $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local/indexes.conf <<'EOF'
[novamart_inv]
homePath    = $SPLUNK_DB/novamart_inv/db
coldPath    = $SPLUNK_DB/novamart_inv/coldb
thawedPath  = $SPLUNK_DB/novamart_inv/thaweddb
EOF
```

2. DEFINE THE SOURCE TYPE: EVENT BREAKING AND TIMESTAMP

```
cat > $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local/props.conf <<'EOF'
[novamart:inventory]
SHOULD_LINEMERGE = false
LINE_BREAKER      = ([\r\n]+)
TIME_PREFIX       = ts=
TIME_FORMAT        = %d-%m-%Y %H:%M:%S.%3N
MAX_TIMESTAMP_LOOKAHEAD = 25
TZ                = Asia/Kolkata
KV_MODE           = auto
EOF
```

3. SET THE HOST ON THE INPUT (APPENDS TO THE MONITOR STANZA, THE LAST ONE IN THE FILE)

```
echo "host = wh-mum01" >> $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local/inputs.conf
```

4. VERIFY WHAT SPLUNK WILL ACTUALLY USE

```
$SPLUNK_HOME/bin/splunk btool inputs list --debug | grep -A5 inventory_wh
$SPLUNK_HOME/bin/splunk btool props list novamart:inventory --debug
$SPLUNK_HOME/bin/splunk btool indexes list novamart_inv
$SPLUNK_HOME/bin/splunk btool check
```

EXPECTED

btool shows each setting prefixed with `.../apps/nm_inventory_inputs/local/...: index = novamart_inv`, `host = wh-mum01`, `sourcetype = novamart:inventory`, and the `TIME_PREFIX / TIME_FORMAT` settings. `btool check` reports no problems for these files.

Part E — Re-read the file: reset the fishbucket

Splunk already read `inventory_wh.log` in Part A, and dropped the events because the index was missing. The **fishbucket** remembers the file as "done", so a restart alone will *not* re-read it. Reset that one file's record:

```
$SPLUNK_HOME/bin/splunk stop
$SPLUNK_HOME/bin/splunk cmd btprobe -d $SPLUNK_HOME/var/lib/splunk/fishbucket/splunk_private_db \
  --file $SPLUNK_HOME/etc/apps/novamart_lab/data/troubleshoot/inventory_wh.log --reset
# Option B: --file /home/ubuntu/novamart_dataset/troubleshoot/inventory_wh.log
$SPLUNK_HOME/bin/splunk start
```

The `--file` path must be exactly the path the monitor uses. Remember to restart the live generator from Lab 12 if it stopped (`nohup python3 novamart_live.py --mode normal --minutes 480 &`).

Part F — Validate

#	Search (All time)	Expected
a	<code> tstats count where index=novamart_inv by host, sourcetype</code>	wh-mum01 · novamart:inventory · 630
b	<code>index=novamart_inv stats min(_time) AS first, max(_time) AS last eval first=strftime(first,"%d-%b %H:%M:%S"), last=strftime(last,"%d-%b %H:%M:%S")</code>	14-Sep 06:05:57 → 20-Sep 22:49:29
c	<code>index=novamart_inv timechart span=1d count</code>	75 per day, except 18-Sep = 180
d	<code>index=novamart_inv stats count by event</code>	STOCK_OUT 336 · STOCK_IN 123 · STOCK_ADJUST 99 · LOW_STOCK 72
e	<code>index=novamart_inv event=STOCK_OUT earliest="09/18/2026:19:00:00" latest="09/19/2026:00:00:00" stats count</code>	64 stock movements out during the sale evening
f	<code>index=novamart_inv event=LOW_STOCK top limit=3 sku</code>	NM-1001 (9) · NM-6002 (8) · NM-1002 (7)
g	<code>index=novamart_inv event=STOCK_ADJUST stats count by operator</code>	arjun.k 38 · priya.n 35 · svc_inventory 26

ROOT-CAUSE SUMMARY

Write one line per fault: symptom → evidence → fix.

CHECKPOINT

- ☐ You found the missing index in Messages and `_internal`.
- ☐ You found the wrong timestamp in the preview without indexing bad data.
- ☐ All fixes live in one app, and btool confirms them.
- ☐ 630 events with correct times and host after the fishbucket reset.

Administration Essentials

Goal: explore the configuration layout, set retention on an index, create a restricted role and user and prove the restrictions work, and use the Monitoring Console and `_internal`.

Part A — Explore \$SPLUNK_HOME

```
ls $SPLUNK_HOME/etc/apps/
ls $SPLUNK_HOME/etc/apps/search/local/      # what you created in Splunk Web on Day 1/2
ls $SPLUNK_HOME/etc/apps/search/lookups/    # uploaded CSVs (region_targets.csv, nm_admin_ips.csv
...)
ls $SPLUNK_HOME/etc/users/admin/search/local/ # your private knowledge objects
grep -A3 "NM - Report - Daily Revenue" $SPLUNK_HOME/etc/apps/search/local/savedsearches.conf
ls $SPLUNK_HOME/var/lib/splunk/ | grep novamart # index directories
tail -5 $SPLUNK_HOME/var/log/splunk/splunkd.log
```

QUESTIONS

(1) Where is the report from Lab 11 stored, and why is it in that directory? (2) Which directories hold the `novamart` and `novamart_inv` index data?

Part B — Retention for novamart_inv

Requirement: keep warehouse data for **1 year**, capped at **5 GB**.

```
cat >> $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local/indexes.conf <<'EOF'
frozenTimePeriodInSecs = 31536000
maxTotalDataSizeMB = 5000
EOF
cat $SPLUNK_HOME/etc/apps/nm_inventory_inputs/local/indexes.conf
$SPLUNK_HOME/bin/splunk btool indexes list novamart_inv | grep -E "frozenTime|maxTotalDataSize"
$SPLUNK_HOME/bin/splunk restart
```

```
| rest /services/data/indexes/novamart_inv
| table title, frozenTimePeriodInSecs, maxTotalDataSizeMB, currentDBSizeMB, totalEventCount
```

EXPECTED

`frozenTimePeriodInSecs=31536000`, `maxTotalDataSizeMB=5000`, `totalEventCount=630`. The two lines are appended to the end of the file, so they belong to the `[novamart_inv]` stanza.

WATCH OUT

Do **not** set a short retention (e.g. 30 days) on `novamart` or `novamart_inv` in the lab. The events date from September 2026, and older buckets would be frozen, which means **deleted**.

Part C — Create a restricted role and a user

1 *Settings* → *Roles* → *New Role*. Name `nm_analyst`. **Inheritance:** tick `user`.

- 2 **Indexes** tab: tick `novamart` as both *Included* and *Default*. Make sure `novamart_inv`, `main` and the internal indexes are **not** included (untick "All non-internal indexes" if it is ticked).
- 3 **Restrictions** tab: search filter `NOT sourcetype=linux_secure`.
- 4 **Resources** tab: role search job limit `5`. **Create**.
- 5 **Settings** → **Users** → **New User**: Name `analyst1`, Full name *NovaMart Analyst*, password of your choice, Time zone *Asia/Kolkata*, Default app *search*. Assign role `nm_analyst` **only** (remove `user` if it was added). Untick "Require password change on first login" for the lab. **Save**.

WHAT SPLUNK WEB WROTE

```
$SPLUNK_HOME/bin/splunk btool authorize list role_nm_analyst | grep -E "importRoles|srchIndexes|srchFilter|srchJobsQuota"
```

Part D — Prove the restrictions

Open a **private/incognito** browser window and log in as `analyst1`.

#	As analyst1, run (All time)	Expected
a	<code>index=novamart stats count by sourcetype</code>	access_combined, novamart:app, novamart:payment (and novamart:firewall if loaded). No linux_secure .
b	<code>index=novamart sourcetype=linux_secure</code>	0 events: the search filter wins
c	<code>index=novamart_inv</code> and <code>index=_internal</code>	No results (not permitted)
d	<code>status=503 stats count</code> (no index)	35, plus any live incident events: <code>novamart</code> is the default index
e	Open Reports	Sees <code>NM - Report - Payment Gateway Performance</code> (shared, App). Does not see <code>NM - Report - Daily Revenue by Category</code> (private).

THINK ABOUT IT

As admin, share a report of `index=novamart sourcetype=linux_secure | stats count by host` with Run As **Owner** and Read = Everyone, then open it as analyst1. What do you see, and why is this a governance risk? (Delete the report afterwards.)

Part E — Monitoring Console tour

As admin: **Settings** → **Monitoring Console**.

Go to	Find and write down
Overview	The instance's role(s) and overall health status
Indexing → Performance → Indexing Performance: Instance	Current indexing rate (KB/s), and whether any queue shows fill above 0%
Indexing → Indexes and Volumes → Index Detail: Instance → <code>novamart_inv</code>	Size on disk, event count, retention settings from Part B

Go to	Find and write down
Search → Scheduler Activity: Instance	Your Lab 11/13 scheduled searches, and whether any were skipped
Settings → License Usage (or Indexing → License Usage)	Today's usage in MB, and the top source type
Health Check → Start	Any warnings (on a lab box you may see ones about resources or default passwords)

Part F — Admin searches in `_internal` and `_audit`

LICENCE USED TODAY BY SOURCE TYPE (MB)

```
index=_internal source=*license_usage.log type=Usage earliest=@d
| stats sum(b) AS bytes by st | eval MB=round(bytes/1024/1024, 3) | sort - MB
```

TOP WARNING/ERROR COMPONENTS IN THE LAST HOUR

```
index=_internal sourcetype=splunkd log_level IN (ERROR, WARN) earliest=-60m
| stats count by component, log_level | sort - count | head 10
```

THROUGHPUT PER SOURCE TYPE (TODAY)

```
index=_internal source=*metrics.log group=per_sourcetype_thruput earliest=@d
| stats sum(kb) AS kb by series | sort - kb
```

WHO LOGGED IN TO SPLUNK TODAY?

```
index=_audit action="login attempt" earliest=@d | stats count by user, info
```

EXPECTED

Licence: `access_combined` (live web03 traffic) is usually the top source type today;
`novamart:inventory` appears after Lab 14 (about 0.07 MB). `_audit` shows successful logins for
`admin` and `analyst1`, plus any failed attempts.

Wrap-up — stop the lab processes

```
kill -f novamart_live.py          # stop the generators
kill -f webhook_listener.py      # stop the webhook listener (or Ctrl+C in Terminal 2)
```

Also make sure both Lab 13 alerts are disabled, and change the Lab 11 report back to a daily schedule (or disable it).

CHECKPOINT

- ☐ You can find apps, local config, lookups, user objects and index data under `$SPLUNK_HOME`.
- ☐ `novamart_inv` has 1-year retention with a 5 GB cap.
- ☐ `analyst1` cannot see `linux_secure`, `novamart_inv` or `_internal`.
- ☐ You have used four Monitoring Console views and four admin searches.

Challenge Exercises

For fast finishers. Solutions are in Appendix A.

#	Challenge	Task
C1	Chart overlay	One chart for 18-Sep 19:00–24:00 at 5-minute resolution: total checkouts as columns and checkout 5xx as a line on a second axis.
C2	Unusual orders	Use <code>eventstats</code> to list successful orders larger than 3 × the average order value. How many are there?
C3	Target tracker	Add a dashboard panel that shows each region's attainment % as a table with colour ranges (red < 90, amber 90–100, green ≥ 100).
C4	Dynamic dropdown	Make the Region dropdown in your dashboard populate itself from the data (<code>stats count by region</code>) instead of static values.
C5	Cause alert	Create an alert on <code>"Circuit breaker" gw-beta</code> WARN events. Backtest it: when would it have fired on 18-Sep?
C6	Index-time filter	Using <code>props.conf</code> / <code>transforms.conf</code> in a new app, drop <code>GET /health</code> requests from new <code>access_combined</code> data. Where does the configuration go on a single instance? How can you prove it works with the live generator? (The live generator does not create health checks, so what would you test with?)
C7	Weekly history	Change the Lab 11 scheduled report so that it <i>appends</i> to <code>nm_daily_revenue.csv</code> and adds a <code>run_time</code> column (<code>eval run_time=now()</code>). Why might append create duplicates?
C8	Admin audit	Which saved searches are scheduled on this instance, who owns them, and how often do they run? (<code> rest /servicesNS/-/-/saved/searches</code>)

Appendix A: Answers and Solutions

Lab 8

8.1 The same visitor usually appears on both web servers (the load balancer spreads requests), so the distinct count over both hosts (425) is far less than the sum of the per-host counts. **8.8d** In `where`, unquoted `East` is a field name, and no field called East exists, so `region=East` is false for every row.

Lab 9: model SRE summary

1. Friday 18-Sep, 20:07–20:45: 35 failed checkouts (HTTP 503), checkout success rate that day 87.7% (normally 92–98%), CARD and NETBANKING most affected.
2. Payment gateway gw-beta: its 5-minute average latency went from ≈650 ms to ≈30,000 ms from 20:05 to 20:40 and recovered after the 20:45 failover. Its median over the week stayed at ≈411 ms, so this was an acute incident rather than a chronic one.
3. `search-service` logs Elasticsearch connection-reset errors every day (38 in the week, 38 HTTP 500s on /search). It needs a ticket for the search team.
4. A 4xx spike on 16-Sep (191 in the day, 153 × 404 from one IP) is a vulnerability scanner; this is also visible in the web error trend.
5. For example, checkout 5xx count per 5 minutes (a direct customer-impact signal), or gw-beta p95 latency (an early warning).

Lab 10

10.6 Think about it. The subsearch returns `(( clientip="10.20.0.5" ) OR ( clientip="192.0.2.66" ) OR ... )`, and the main search becomes `... NOT ( ... )`. You can see this in `normalizedSearch` (the exact formatting varies).

Lab 11

11.4 Think about it. No. With Run As Owner, the report runs with the owner's access (e.g. admin), so every reader would see security events that their own roles forbid. Use Run As **User**, or restrict read access to security roles. Lab 15 Part D demonstrates this.

Lab 14: root-cause summary

Fault	Symptom	Evidence	Fix
1. Missing index	No data in <code>index=novamart_inv</code>	Messages + splunkd.log: <i>received event for unconfigured/disabled/deleted index</i>	<code>[novamart_inv]</code> in <code>indexes.conf</code>
2. Wrong host	host would be the Splunk server's name	Same warning, <code>host:: <server></code> ; no <code>host</code> in btool output	<code>host = wh-mum01</code> in <code>inputs.conf</code>
3. Wrong timestamp	Events dated August 2026	Add Data preview: <code>_time = batch_date</code>	<code>TIME_PREFIX = ts=</code> , <code>TIME_FORMAT</code> , <code>TZ</code> in <code>props.conf</code>

Fault	Symptom	Evidence	Fix
Re-index	Still no data after fixing	File already recorded in the fishbucket	<code>btprobe ... --reset</code> with Splunk stopped

Lab 15

Part A. (1) In `etc/apps/search/local/savedsearches.conf`, because it was created in the Search app and then shared at app level. While it was private, it lived under `etc/users/admin/search/local/`. (2) `var/lib/splunk/novamart/` and `var/lib/splunk/novamart_inv/` (each with `db`, `colddb`, `thaweddb`).

Part D, Think about it. analyst1 sees the linux_secure counts: the report ran as its owner (admin), which bypasses the analyst's search filter. That is why Run As matters.

Challenge solutions

C1 OVERLAY

```
index=novamart sourcetype=access_combined uri_path=/checkout earliest="09/18/2026:19:00:00" latest="09/19/2026:00:00:00"
| timechart span=5m count AS checkouts, count(eval(status≥500)) AS errors
''' Visualization: Column. Format → Chart Overlay → Overlay: errors, View as Axis: On '''
```

C2 UNUSUAL ORDERS

```
index=novamart sourcetype=novamart:app action=checkout status=SUCCESS
| eventstats avg(amount) AS aov
| where amount > 3*aov
| stats count, values(product_id) AS products
```

The average order value is ₹6,394.70, so the threshold is ₹19,184. Result: **47** orders: 24 × one TV (NM-1004), 5 × two TVs, 11 × two smartphones (NM-1001), and 7 × three smartphones.

C3 TARGET TRACKER

```
index=novamart sourcetype=novamart:app action=checkout status=SUCCESS
| stats sum(amount) AS revenue by region
| lookup region_targets region OUTPUT weekly_revenue_target_inr AS target
| eval attainment_pct=round(revenue/target*100, 1)
| table region, revenue, target, attainment_pct
```

C4 DYNAMIC DROPDOWN SEARCH (STUDIO: DROPDOWN → DYNAMIC OPTIONS → DATA SOURCE)

```
index=novamart sourcetype=novamart:app action=checkout | stats count by region | sort region
```

C5 CAUSE ALERT BACKTEST

```
index=novamart sourcetype=novamart:app "Circuit breaker" gw-beta earliest="09/18/2026:00:00:00" latest="09/19/2026:00:00:00"
| table _time, _raw
```

The first event is at 20:05:01, so a 1-minute alert fires at about 20:06, more than a minute before the first customer error (20:07:38).

```

C6 PROPS.CONF + TRANSFORMS.CONF IN $SPLUNK_HOME/ETC/APPS/NM_FILTERS/LOCAL/ (SINGLE INSTANCE = INDEX
ER)

[access_combined]
TRANSFORMS-drop_health = nm_drop_health

[nm_drop_health]
REGEX = "GET /health HTTP
DEST_KEY = queue
FORMAT = nullQueue

```

The first stanza goes in `props.conf`, the second in `transforms.conf`; then restart. Test by appending a fake health-check line to `web_live_access.log` (`echo '10.20.0.5 - - [<now>] "GET /health HTTP/1.1" 200 17 "-" "ELB-HealthChecker/2.0" ' >> ...`) and confirm that it never appears, while other new events do. Historical events are unaffected.

```

C8 SCHEDULED SEARCHES
| rest /servicesNS/-/-/saved/searches
| search is_scheduled=1
| table title, eai:acl.owner, eai:acl.app, cron_schedule, disabled, next_scheduled_time

```

Appendix B: Lab Troubleshooting

Problem	Fix
Historical counts are higher than this guide says	Live data has been added. Use <i>Date Range 14–20 Sep 2026</i> or <code>earliest=0 latest="09/21/2026:00:00:00"</code> .
No <code>host=web03</code> events	Is the generator running (<code>pgrep -af novamart_live</code>)? Option A: is the app at v1.1 (the live inputs are in 1.1)? Option B: did you add the three monitors (<code>splunk list monitor grep live</code>)?
Live SSH events have the wrong time	Option B: the <code>linux_secure</code> source type must have <code>TZ = Asia/Kolkata</code> (Day 1 Lab 2 Part D). Option A: the app sets it for you.
<code>lookup</code> error: "could not find all of the specified lookup fields"	Check the column names with <code> inputlookup <name></code> ; check the definition and file permissions (share to App / Everyone).
CIDR lookup returns nothing	Set Match type <code>CIDR(cidr)</code> in the lookup definition's advanced options.
Dashboard panel "Waiting for input"	The input has no default value, or the token name in the SPL does not match the input's token name exactly (case-sensitive).
Scheduled report never ran	Is <i>Schedule Report</i> ticked? Check <code>index=_internal sourcetype=scheduler savedsearch_name="..."</code> . Wait for the next cron minute.
Alert fires but no webhook	Is <code>webhook_listener.py</code> running on the Splunk server (port 8765)? Check <code>component=sendmodalert</code> . If Splunk enforces a webhook allow list, add <code>http://127.0.0.1:8765/*</code> .
Alert never fires	Run the alert search manually during the incident. Check the cron schedule, that the alert is enabled, the time range, and whether it is still throttled from an earlier trigger.
Lab 14: still no data after the fixes	Did you stop Splunk before <code>btprobe</code> , and use exactly the monitored path in <code>--file</code> ? Check <code>splunk list inputstatus grep -A3 inventory</code> .
Lab 15: analyst1 sees <code>linux_secure</code>	Check the role's search filter, and that analyst1 does not also have the <code>user</code> role (roles combine, and a less restrictive role can widen access).

Appendix C: NovaMart Operations Dashboard (Classic Simple XML)

Create a **Classic** dashboard (*Dashboards* → *Create New Dashboard* → *Classic Dashboards*), open **Edit** → **Source**, replace everything with the XML below, and **Save**. It contains the same panels, inputs, tokens and drilldown as Lab 12. The file is also supplied as `tools/novamart_operations_classic.xml` in the dataset.

```
<form version="1.1" theme="light">
  <label>NovaMart Operations (Classic)</label>
  <description>Platform health, payments and business KPIs - Splunk Fundamentals Day 2, Lab 12</description>
  <fieldset submitButton="false" autoRun="true">
    <input type="time" token="t" searchWhenChanged="true">
      <label>Time range</label>
      <default>
        <earliest>1789324200</earliest>
        <latest>1789929000</latest>
      </default>
    </input>
    <input type="dropdown" token="host" searchWhenChanged="true">
      <label>Host</label>
      <choice value="*">All</choice>
      <choice value="web01">web01</choice>
      <choice value="web02">web02</choice>
      <choice value="web03">web03 (live)</choice>
      <default>*</default>
    </input>
    <input type="dropdown" token="region" searchWhenChanged="true">
      <label>Region</label>
      <choice value="*">All</choice>
      <choice value="North">North</choice>
      <choice value="South">South</choice>
      <choice value="East">East</choice>
      <choice value="West">West</choice>
      <default>*</default>
    </input>
  </fieldset>
  <row>
    <panel>
      <single>
        <title>Requests</title>
        <search>
          <query><![CDATA[index=novamart sourcetype=access_combined host=$host$ | stats count]]></query>
          <earliest>$t.earliest$</earliest>
          <latest>$t.latest$</latest>
        </search>
        <option name="numberPrecision">0</option>
        <option name="useThousandSeparators">1</option>
      </single>
    </panel>
    <panel>
      <single>
        <title>Server errors (5xx)</title>
        <search>
          <query><![CDATA[index=novamart sourcetype=access_combined host=$host$ status>=500 | stats count]]></query>
          <earliest>$t.earliest$</earliest>
          <latest>$t.latest$</latest>
        </search>
        <option name="colorBy">value</option>
        <option name="colorMode">block</option>
        <option name="rangeColors">["0x53a051", "0xf8be34", "0xdc4e41"]</option>
        <option name="rangeValues">[4,19]</option>
        <option name="useColors">1</option>
      </single>
    </panel>
    <panel>
      <single>
        <title>Successful orders</title>
        <search>
          <query><![CDATA[index=novamart sourcetype=novamart:app action=checkout status=SUCCESS region=$region$ | stats count]]></query>
          <earliest>$t.earliest$</earliest>
          <latest>$t.latest$</latest>
        </search>
      </single>
    </panel>
  </row>
</form>
```

```

<title>Revenue (INR)</title>
<search>
  <query><![CDATA[index=novamart sourcetype=novamart:app action=checkout status=SUCCESS region=$region$ | stats sum(amo
unt) AS revenue]]></query>
  <earliest>$t.earliest$</earliest>
  <latest>$t.latest$</latest>
</search>
<option name="numberPrecision">0</option>
<option name="unit">₹</option>
<option name="unitPosition">before</option>
<option name="useThousandSeparators">1</option>
</single>
</panel>
</row>
<row>
  <panel>
    <chart>
      <title>Requests over time</title>
      <search>
        <query><![CDATA[index=novamart sourcetype=access_combined host=$host$ | timechart count by host]]></query>
        <earliest>$t.earliest$</earliest>
        <latest>$t.latest$</latest>
      </search>
      <option name="charting.chart">line</option>
      <option name="charting.legend.placement">bottom</option>
    </chart>
  </panel>
  <panel>
    <chart>
      <title>Server errors over time</title>
      <search>
        <query><![CDATA[index=novamart sourcetype=access_combined host=$host$ status>=500 | timechart count by status]]></que
ry>
        <earliest>$t.earliest$</earliest>
        <latest>$t.latest$</latest>
      </search>
      <option name="charting.chart">column</option>
      <option name="charting.chart.stackMode">stacked</option>
      <option name="charting.legend.placement">bottom</option>
    </chart>
  </panel>
</row>
<row>
  <panel>
    <table>
      <title>Errors by status (click a row)</title>
      <search>
        <query><![CDATA[index=novamart sourcetype=access_combined host=$host$ status>=400 | stats count by status | sort - co
unt]]></query>
        <earliest>$t.earliest$</earliest>
        <latest>$t.latest$</latest>
      </search>
      <option name="drilldown">row</option>
      <drilldown>
        <set token="sel_status">$row.status$</set>
      </drilldown>
    </table>
  </panel>
  <panel depends="$sel_status$">
    <table>
      <title>Events for status $sel_status$</title>
      <search>
        <query><![CDATA[index=novamart sourcetype=access_combined host=$host$ status=$sel_status$ | table _time, host, client
ip, method, uri_path, status, useragent | head 100]]></query>
        <earliest>$t.earliest$</earliest>
        <latest>$t.latest$</latest>
      </search>
      <option name="count">10</option>
      <option name="drilldown">none</option>
    </table>
  </panel>
</row>
<row>
  <panel>
    <chart>
      <title>Revenue by category</title>
      <search>
        <query><![CDATA[index=novamart sourcetype=novamart:app action=checkout status=SUCCESS region=$region$ | lookup novama
rt_products product_id OUTPUT category | stats sum(amount) AS revenue by category | sort - revenue]]></query>
        <earliest>$t.earliest$</earliest>
        <latest>$t.latest$</latest>
      </search>
      <option name="charting.chart">bar</option>
      <option name="charting.legend.placement">none</option>

```

```

    </chart>
  </panel>
</panel>
<panel>
  <chart>
    <title>Payment latency by gateway (ms)</title>
    <search>
      <query><![CDATA[index=novamart sourcetype=novamart:payment | timechart avg(latency_ms) by gateway]]></query>
      <earliest>${t.earliest}</earliest>
      <latest>${t.latest}</latest>
    </search>
    <option name="charting.chart">line</option>
    <option name="charting.legend.placement">bottom</option>
  </chart>
</panel>
</panel>
<panel>
  <chart>
    <title>Live: requests per minute (web03, last 15 min)</title>
    <search>
      <query><![CDATA[index=novamart sourcetype=access_combined host=web03 | timechart span=1m count]]></query>
      <earliest>-15m</earliest>
      <latest>now</latest>
      <refresh>60s</refresh>
      <refreshType>delay</refreshType>
    </search>
    <option name="charting.chart">line</option>
    <option name="charting.legend.placement">none</option>
  </chart>
</panel>
</row>
</form>

```

Appendix D: Expected Results Card (14–20 Sep 2026)

Check	Value	Check	Value
Requests web01 / web02	10,856 / 11,785	Revenue / orders	5,192,498 / 812
Distinct client IPs	425	Attainment N / S / W / E	118.1 / 107.3 / 90.6 / 88.4 %
5xx web01 / web02	43 / 30	Accepted SSH from Internet	1 (deploy, password)
Busiest hour of week	21:00 (3,039)	Admin IPs (outputlookup)	4
Checkout availability 18-Sep	92.71% (480 / 35 errors)	Checkout alert windows ≥ 5	20:15, 20:20, 20:30, 20:40
Checkout success rate 18-Sep	87.7%	Brute-force windows > 20	02:10 - 02:30 (5)
gw-beta avg latency 18-Sep	4,731.6 ms	novamart_inv events	630 (18-Sep: 180)
Human (non-bot) requests	21,371	novamart_inv first / last	14-Sep 06:05:57 / 20-Sep 22:49:29
search-service errors (week)	38	gw-alpha / gw-beta p95	≈ 812 / $\approx 30,260$ ms